

Bio mining (Mineral leaching) - It is technique of using prokaryotes such as Bacteria / fungi to extract Metals from their ores. The ability of M.O. to solubilize Metals from Insoluble Metals is called "Bioleaching".

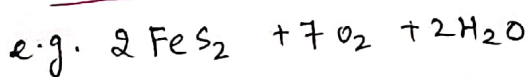
Microorganism for Bioleaching

- ⇒ Most commonly used M.O.
 - Thiobacillus ferrooxidans
 - Thiobacillus thiooxidans
- ⇒ Sulfolobus acidocaldarius, Leptospirillum ferrooxidans
- Thiobacillus organoparvus, Pseudomonas aeruginosa
- ⇒ Certain fungi *Aspergillus niger* (to extract Cu, Ni)
- Aspergillus oryzae* (for gold)

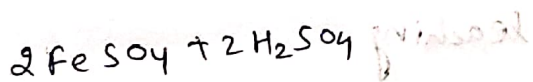
Mechanism of Bio mining

Direct

- In this process, there is a direct enzymatic attack on the minerals by M.O.
- M.O. can oxidize metal sulfides by a direct mechanism obtaining e^- directly from reduced minerals.

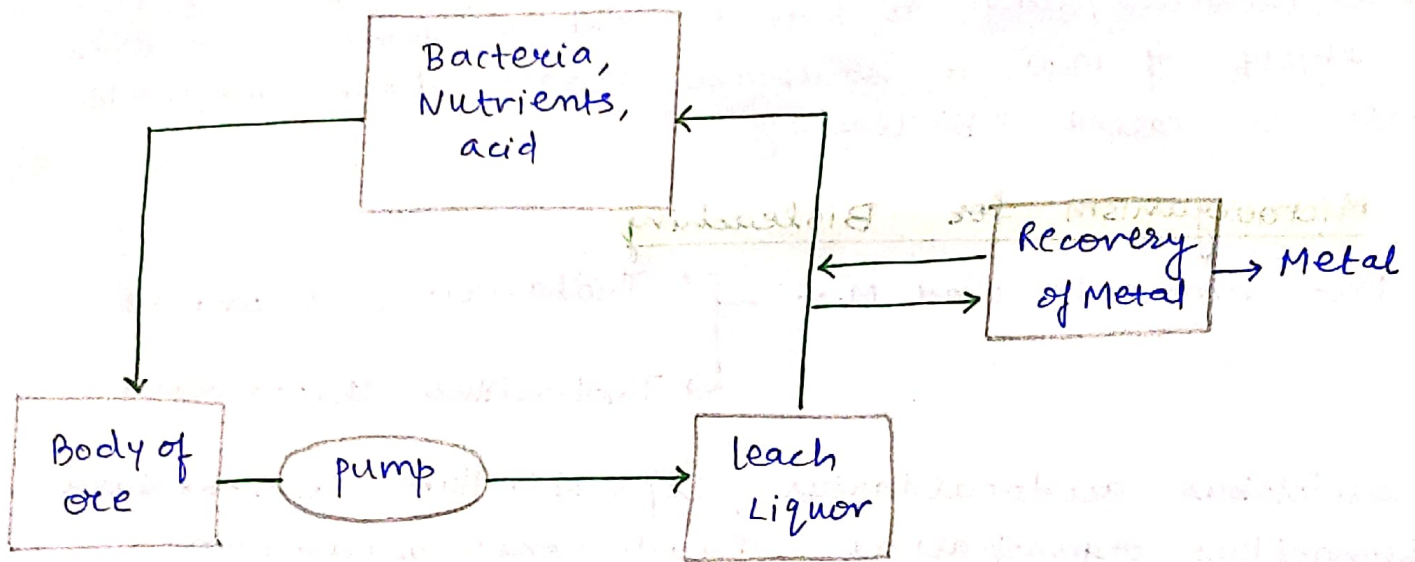


↓ Thiobacillus



Indirect Method.

- No need of direct physical contact b/w M.O. & Mineral ore.
- Iron act as a e^- carrier. It obtain e^- from Mineral & transfer to M.O.
- Bacteria produce strong oxidizing agents such as ferric iron, sulfuric acid on oxidation of soluble Fe/S
- Ferric / Iron / sulfuric acid being powerful oxidizing agents react with mineral metals & extract them.



A flow Diagram for Microbial Bioleaching

Commercial process of Bioleaching

- ⇒ The Naturally occurring Mineral leaching is very slow.
- ⇒ Microbial Bioleaching process can be optimized by creating ideal conditions (temp, pH, Nutrient, O_2 , CO_2)
- ⇒ The desired M.O. with Nutrients, acid etc. are pumped into the ore Bed. The M.O. grow & produce more acid.
- ⇒ The extracted leach liquor is processed for Metal Recovery. The leach liquor can be recycled again & again for further Metal Extraction.

3 Methods used

slope
leaching

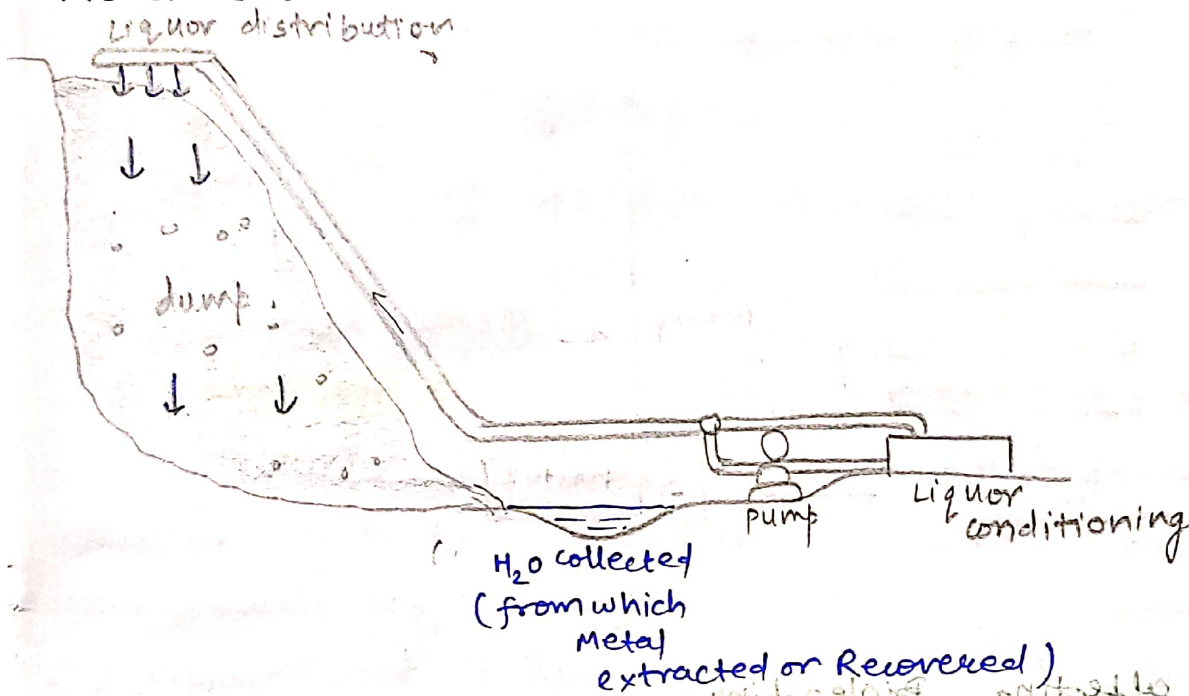
Heap
leaching

In-situ
leaching

(1) slope leaching

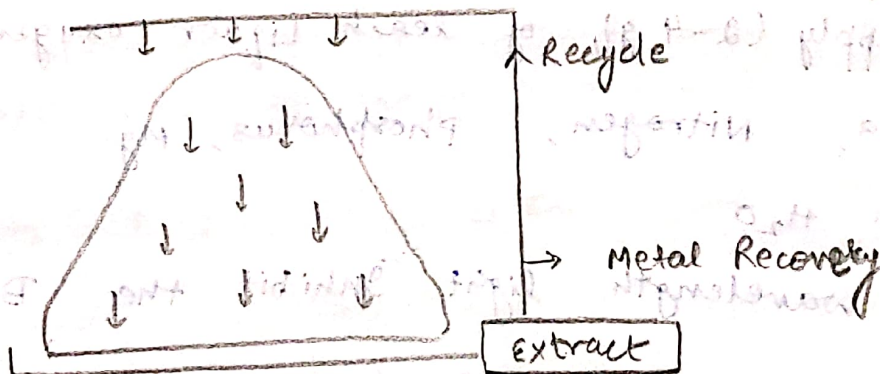
⇒ finely ground ores are dumped in large piles down a mountain side & continuously sprinkled with water containing *Thiobacillus*.

⇒ The H_2O is collected at the bottom and reused after Metal Extraction.



(2) Heap leaching

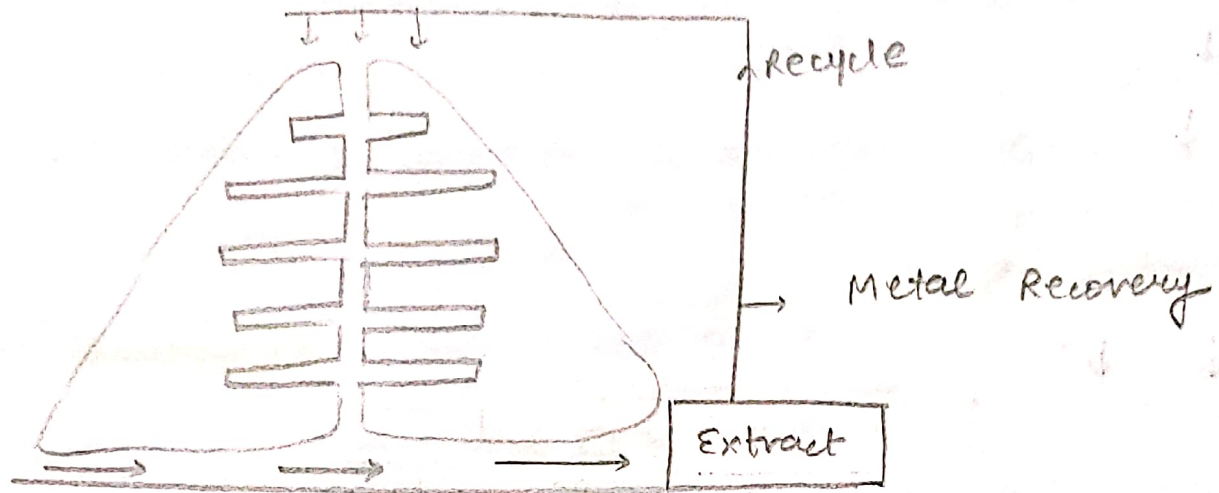
- The ore is arranged in form of large Heaps.
- Rest process same as in slope leaching.



(3) In-situ leaching

⇒ Water containing Thiobacillus is pumped through drilled passages to unextracted ore which remains in its original location in the earth.

⇒ The Acidic H_2O seeps through the rock and collect in the Bottom most cavity from which it is pumped, the minerals extracted, and the H_2O reused after regeneration of Bacteria.



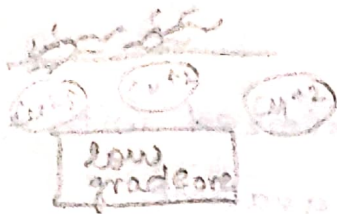
Factors affecting Bioleaching

e.g. For *Thiobacillus ferrooxidans*.

- (1) Temp. (Optimum 30 to 50°C)
- (2) pH (Optimum 2.2 to 2.5)
- (3) Iron supply (2-4 g/L of leach liquor oxygen)
- (4) Ammonia, Nitrogen, Phosphorus, Mg
- (5) optimum H_2O
- (6) short wavelength light inhibit the Bacteria.

Copper Bioleaching

Bacteria are grown on low grade ores



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Biological & chemical processes
of Bacteria form a solution
rich in Cu ions (leachate)

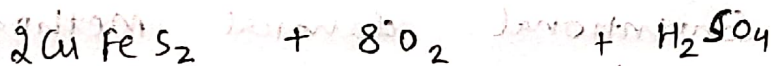
Cu²⁺ leachate

↓
Copper containing solution is removed, the Cu is precipitated

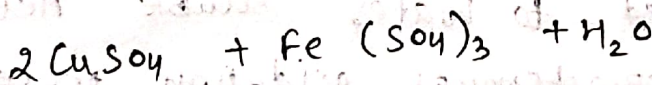
e.g. **Chalcopyrite** (contains 26% Cu, 26% Fe, 33% S, 2.5% Zn)

- This is carried out by M.O. *Thiobacillus ferrooxidans* which oxidises insoluble chalcopyrite (CuFeS_2) and converts it into soluble copper sulfate (CuSO_4)
- sulphuric acid, a byproduct formed in this reaction, maintains acidic env. (low pH) required for growth of M.O.
- Copper leaching is usually carried out by heap and in situ process.

⇒ As the copper containing solution (i.e. copper in dissolved state) comes out, copper can be precipitated and the water is recycled after adjusting the pH to around 2.



↓ *Thiobacillus*



Advantages

- (1) Bioleaching can recover metals from low grade ores
- (2) Cost effective
- (3) It can be successfully employed for concentrating Metals from wastes or dilute mixture.
- (4) Eco-friendly (since it does not cause any kind of pollution which was the problem in case of conventional mining techniques)
- (5) It can be used to produce refined and expensive Metals which otherwise may not be possible.
- (6) Simple process with low cost technology.
- (7) It is ideally suited for the developing countries.

Limitations

- (1) Time consuming
- (2) Very low yield of Minerals
- (3) Requires a large open area for treatment
- (4) ↑ risk of contamination
- (5) Inconsistent yield because Bacteria cannot grow uniformly